

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Computer Vision with OpenCV COURSE CODE: DSA02

COURSE OUTCOME COVERED

CO2: Apply image processing and feature extraction techniques, including edge detection, motion estimation, and optical flow, for analyzing images.. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 5 Total Marks:50 Annexure A

Analytical Problem Solving

Code of Conduct:

I M Tilak Venkata Sai(192424415)___ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION

Numerical Problems						
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	

Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	
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Annexure A

Analytical Problem Solving

Question 1: Salt-and-Pepper Noise Removal Before Edge Detection Scenario: A grayscale image of size 1024×1024 is captured using a surveillance camera under poor lighting conditions. The image is corrupted by salt-and-pepper noise with a noise density of 0.05. This means that some pixels are randomly changed to either black or white intensity values. The image has to be preprocessed before applying an edge detection algorithm. Since edge detection is sensitive to abrupt intensity variations, the selected filter should remove noise effectively while preserving important edge details.

Answer the following:

- Identify the most suitable filtering technique for removing salt-and-pepper noise without causing significant edge blurring.
- Explain why the selected filter performs better than an averaging filter for impulse noise removal.
- If a 3×3 median filter is applied, estimate the probability that a noisy center pixel will be corrected. Clearly state your assumptions.
- Estimate the total number of noisy pixels present in the original image.
- Estimate the expected number of noisy pixels remaining after median filtering.
- Explain how the preprocessing strategy would change if the same image was affected by Gaussian noise instead of salt-and-pepper noise.

Question 2: Discretization of a Continuous Image

Scenario : An analog grayscale image has a smooth sinusoidal intensity variation along the horizontal direction. The intensity changes continuously with respect to the spatial coordinate x , while it remains constant along the y -direction. The image is defined over a spatial range of 0 to 255 in both directions. This continuous image has to be converted into a digital image for storage and further processing in a computer vision system.

Answer the following:

- Explain how the continuous image can be converted into a digital image using sampling and quantization.
- Identify the spatial variation pattern of the image in the x -direction.
- Determine the minimum sampling requirement needed to avoid aliasing in the x direction.
- Explain what will happen if the image is sampled below the required sampling rate.
- Suggest a suitable 8-bit quantization scheme for storing the image.
- Discuss the effect of quantization on image quality, storage size, and intensity accuracy.

Question 3: Edge Detection in Thin Line Images

Scenario: A 512×512 grayscale image contains very thin vertical and horizontal white lines of 1-pixel thickness on a dark uniform background. Such images are commonly used in document analysis, engineering drawing interpretation, circuit diagram processing, and road-lane detection.

The objective is to detect the line boundaries accurately without losing thin edge information.

Answer the following:

- Compare Sobel, Prewitt, and Canny edge detectors for detecting 1-pixel-thick vertical and horizontal lines.
- For a horizontal edge located near pixel position (256, 256), compute the Sobel gradient response using the standard Sobel operator.
- Explain how the gradient magnitude and gradient direction help in identifying the strength and orientation of an edge.
- Discuss the role of Gaussian smoothing in the Canny edge detection process. e. Explain how Gaussian smoothing may affect edge localization, especially for thin lines.
- If the Canny threshold is reduced to half of its original value, explain the expected changes in detected edges, weak edges, and false edges.

Question 4: Feature Extraction for Handwritten Digit Recognition Scenario: A computer vision system is being developed for recognizing handwritten digits from scanned images. The handwritten digits vary in size, orientation, stroke thickness, and writing style. Therefore, robust feature extraction is required before classification.

Answer the following:

- Explain the working principles of SIFT, SURF, and ORB feature extraction methods.
- Compare SIFT, SURF, and ORB in terms of scale invariance, rotation invariance, computational cost, and real-time suitability.
- If a digit image is rotated by 45 degrees, identify which feature extraction methods can still produce reliable matching features.
- If the average number of keypoints detected per image is 120 and 50 images are processed, compute the total number of keypoints detected.
- Suggest a suitable dimensionality reduction technique to reduce the feature size while retaining 95% of the useful information.
- Explain how dimensionality reduction improves classification efficiency in handwritten digit recognition.

Question 5: Morphological Preprocessing for Character Recognition Scenario: A binary image of size 256×256 contains scanned handwritten characters. Due to poor scanning quality, the image contains small white noise dots in the background and small black gaps inside the character strokes.

Before applying character recognition, the image must be cleaned using morphological preprocessing techniques.

Answer the following:

- Identify the morphological operation suitable for removing small white noise dots

from the background.

b. Identify the morphological operation suitable for filling small gaps or holes inside the character strokes.

c. Explain the difference between opening and closing operations using simple image processing terminology.

d. If a 3×3 square structuring element is used, explain how it affects thin character strokes.

e. Discuss the risk of using a large structuring element during preprocessing. f.

Propose a complete preprocessing pipeline for handwritten character recognition, starting from image acquisition to feature extraction.

Question 1: Salt-and-Pepper Noise Removal Before Edge Detection

a. Most suitable filtering technique

Median Filter (3×3)

- Removes salt-and-pepper (impulse) noise effectively.
- Preserves image edges better than linear filters.
- Widely used before edge detection.

b. Why is Median Filter better than Averaging Filter?

Median Filter

Removes impulse noise effectively

Replaces center pixel with median value

Preserves sharp edges

Best for salt-and-pepper noise

Averaging Filter

Reduces noise but blurs edges

Replaces with average intensity

Smooths and blurs edge boundaries

Better for Gaussian noise

c. Probability that a noisy center pixel is corrected

Given:

- Noise density = **0.05**
- Window size = **3×3 (9 pixels)**

Assumption:

- Majority of neighboring pixels are not noisy.

Probability a neighboring pixel is clean:

$$P(\text{clean}) = 1 - 0.05 = 0.95$$

Expected clean pixels:

$$9 \times 0.95 = 8.55$$

Since most pixels are clean, the noisy center pixel will almost always be replaced by a correct neighboring value.

Estimated Probability $\approx 99\%$ (≈ 0.99)

d. Total noisy pixels

Image size:

$$1024 \times 1024 = 1,048,576 \text{ pixels}$$

Noise density:

$$0.05$$

Noisy pixels:

$$1,048,576 \times 0.05 = 52,428.8$$

$\approx 52,429$ noisy pixels

e. Expected noisy pixels after median filtering

Median filtering removes most impulse noise.

Typical removal efficiency $\approx 95\%$

Remaining noise:

$$52,429 \times 0.05 = 2,621$$

$\approx 2,621$ noisy pixels remain

f. If image has Gaussian noise instead

Median filter is **not preferred**.

Use:

- Gaussian Filter
- Wiener Filter
- Bilateral Filter

Reason:

Gaussian noise affects almost every pixel with small intensity variations rather than isolated pixels.

Question 2: Discretization of a Continuous Image

a. Conversion using Sampling and Quantization

Sampling

- Converts continuous spatial coordinates into pixels.

Quantization

- Converts continuous intensity values into discrete intensity levels.

Digital Image = **Sampling + Quantization**

b. Spatial variation pattern

The intensity changes as a **sinusoidal wave** along the x-direction.

Along y-direction:

- Constant intensity

c. Minimum sampling requirement

According to the **Nyquist Sampling Theorem**

Sampling Frequency

$$f_s \geq 2f$$

Sampling should be **at least twice** the highest spatial frequency.

d. Effect of sampling below Nyquist rate

Aliasing occurs.

Effects:

- Distorted image
- False patterns
- Jagged edges
- Loss of detail

e. Suitable quantization

8-bit quantization

Levels:

$$2^8 = 256$$

Intensity range:

0–255

f. Effect of quantization

Image Quality

- Higher bit depth → better quality

Storage

8-bit image:

1 byte/pixel

Intensity Accuracy

256 gray levels provide good visual quality.

Question 3: Edge Detection in Thin Line Images

a. Comparison

Feature	Sobel	Prewitt	Canny
Noise Handling	Medium	Low	Excellent
Edge Accuracy	Good	Moderate	Excellent
Thin Edge Detection	Moderate	Moderate	Best
Real-time Speed	Fast	Very Fast	Moderate

Best: Canny

b. Sobel Gradient Response

Sobel kernels

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$
$$G_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

For a horizontal edge:

$$G_x \approx 0$$

$$G_y = \text{Maximum}$$

Gradient magnitude

$$G = \sqrt{G_x^2 + G_y^2}$$

$\approx$ **Maximum**

c. Gradient Magnitude & Direction

Gradient Magnitude

- Edge strength

Gradient Direction

- Edge orientation

Example:

- $0^\circ \rightarrow$ Vertical edge
- $90^\circ \rightarrow$ Horizontal edge

d. Role of Gaussian smoothing

Removes noise before edge detection.

Benefits

- Fewer false edges
- Stable edge detection

e. Effect on thin lines

Advantages

- Removes noise

Disadvantages

- Slight edge blur
- Thin lines may become weaker

f. Threshold reduced by half

Effects

- More edges detected
- More weak edges retained
- More false edges/noise detected

Question 4: Feature Extraction for Handwritten Digit Recognition

a. Working Principles

SIFT

- Detects scale-space extrema
- Uses Difference of Gaussian (DoG)

SURF

- Uses Hessian Matrix
- Faster than SIFT

ORB

- Uses FAST keypoint detector
- BRIEF descriptor

b. Comparison

Feature	SIFT	SURF	ORB
Scale Invariant	Yes	Yes	Partial
Rotation Invariant	Yes	Yes	Yes
Speed	Slow	Medium	Fast
Real-time	No	Moderate	Yes

c. Rotated by 45°

Reliable:

- SIFT ✓
- SURF ✓
- ORB ✓

d. Total keypoints

Average = 120

Images = 50

$$120 \times 50 = 6000$$

Answer: 6000 keypoints

e. Suitable dimensionality reduction

Principal Component Analysis (PCA)

Retain:

95% variance

f. Benefits of dimensionality reduction

- Faster training
- Lower memory usage
- Removes redundant features
- Reduces overfitting
- Improves classification speed

Question 5: Morphological Preprocessing

a. Remove white noise dots

Opening

(Erosion → Dilation)

b. Fill black gaps

Closing

(Dilation → Erosion)

c. Difference

Opening

Removes small white objects

Smooths object boundaries

Closing

Fills small black holes

Connects broken regions

d. Effect of 3×3 structuring element

- Removes tiny noise
- Slightly thins narrow strokes
- Preserves most character shapes

e. Risk of large structuring element

- Loss of fine details
- Thin strokes disappear
- Characters merge
- Reduced recognition accuracy

f. Complete preprocessing pipeline

1. **Image Acquisition** (Scanner/Camera)
2. **Grayscale Conversion**
3. **Noise Removal** (Median/Gaussian Filter)
4. **Binarization** (Otsu Thresholding)
5. **Morphological Opening** (Remove white noise)
6. **Morphological Closing** (Fill gaps in strokes)
7. **Resize & Normalize**
8. **Feature Extraction** (SIFT/SURF/ORB or HOG)
9. **Classification** (e.g., SVM, CNN, or KNN)